

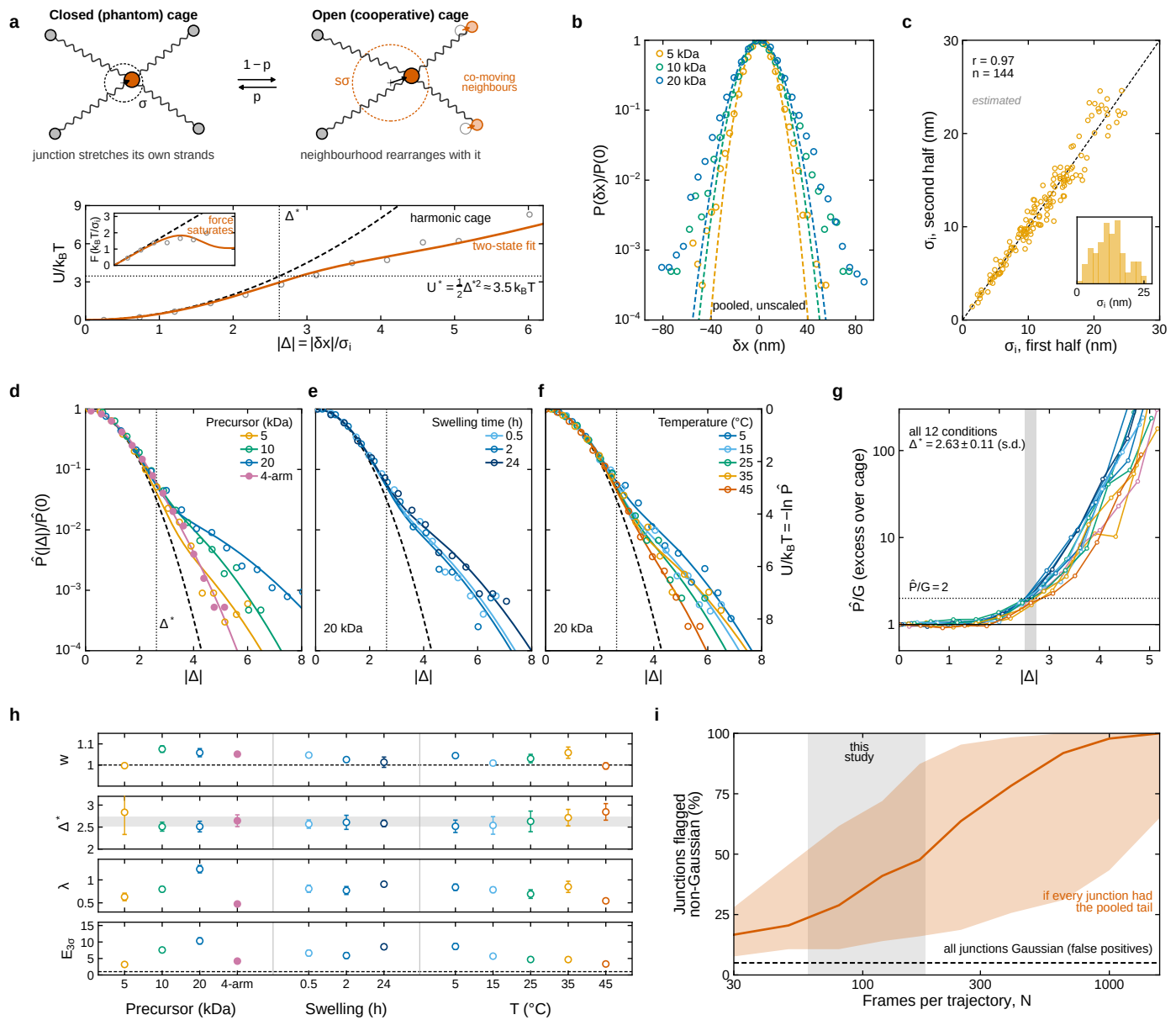


Fig. 5 | Beyond about 2.6σ , junctions leave the harmonic cage together with their neighbourhood. **a**, Two-state cage. Top: in the closed (phantom) cage the junction (vermillion) moves alone and stretches its own strands, with cage width σ . In the open cage, neighbouring junctions move with it (open circles mark their former positions), widening the cage to $s\sigma$. The open state is occupied a fraction p of the time. Bottom: effective potential $U = -k_B T \ln \hat{P}$ for PEGDA 20 kDa swollen for 2 h (grey), with the unit harmonic cage (dashed) and the two-state fit (vermillion). Dotted lines mark Δ^* and $U^* = \frac{1}{2}\Delta^{*2} \approx 3.5 k_B T$. Inset: restoring force $F = dU/d\Delta$, from finite differences of the data (grey) and from the fit; it is linear inside the cage and saturates beyond Δ^* . **b**, Pooled, unscaled distributions of junction displacements δx about their mean positions for three precursor masses (swollen 2 h), with Gaussian fits (dashed; $\sigma = 9.4, 11.7$ and 12.9 nm). Pooling mixes junctions with different cage widths σ_i . **c**, σ_i is a persistent property of each junction: cage widths from the first and second halves of each trajectory (PEGDA 5 kDa, $n = 144$ junctions). Inset: distribution of σ_i . Split-half values are estimated from the measured σ_i with the sampling error of 85 frames per half. **d–f**, Rescaled distributions $\hat{P}(|\Delta|)$, $\Delta = \delta x/\sigma_i$, for precursor mass (**d**; filled symbols, 4-arm PEG-SH 20 kDa), swelling time (**e**; 20 kDa) and temperature (**f**; 20 kDa, swollen 1 h). Dashed: unit Gaussian (no free parameter). Lines: two-state fits, equation (1). Dotted: Δ^* . The right axis of **f** gives $U/k_B T$. **g**, Excess over the harmonic cage, $\hat{P}/G$, for all 12 conditions (colours as in **d–f**). Every condition follows the cage ($\hat{P}/G = 1$) up to $|\Delta| \approx 2$ and doubles at $\Delta^* = 2.63 \pm 0.11$ (grey band; mean $\pm$ s.d., $n = 12$ conditions). **h**, Parameters by condition: w , width of a Gaussian fit to the core ($|\Delta| < 2$); Δ^* ; λ , decay length of an exponential fit to $|\Delta| \geq 2.5$; $E_{3\sigma}$, frequency of $|\Delta| > 3$ relative to the harmonic cage, from the two-state fit. Error bars: s.e. of the fits. The 20 kDa, 2 h sample appears in both the precursor and the swelling series, measured on independent gels. **i**, Fraction of junctions that a Shapiro–Wilk test ($\alpha = 0.05$) flags as non-Gaussian, against trajectory length N . Simulated for Gaussian junctions (dashed) and for junctions that all share the pooled $\hat{P}$ (line: median; band: range over the 12 conditions). Grey band: trajectory lengths in this study (estimated, 60–180 frames). In **b** and **d–g**, x and y components and both signs of Δ are pooled after checking symmetry (Supplementary Fig. X).

Fig. 5: focus, logic and editorial choices

The one message

The phantom cage is exactly harmonic, with no free parameter, up to $\Delta^* \approx 2.6\sigma_i$ ($U^* \approx 3.5 k_B T$) in every network we measured. Beyond that point junctions move together with their neighbourhood. This cooperative mode starts at the same place in every condition and is insensitive to swelling. Its weight is set by strand length and by temperature. Everything in the figure either establishes this claim or protects it against the obvious objection, that the tails come from heterogeneity or from poor statistics.

Logic arc through the panels

Panel	Role	What the reader takes away
a	Concept first	Two cages and one landscape: harmonic up to U^* , then much softer. The measured points on the landscape tie the cartoon to data from the start. The force inset gives the mechanical reading (the cage yields: the force saturates) without a separate panel.
b	Raw observation	The pooled distributions have tails, but pooling mixes junctions with different σ_i . This is the confound to remove.
c	Why rescaling is legitimate	σ_i belongs to the junction: it is reproducible within a trajectory and varies between junctions. Dividing by it removes a static property and nothing else.
d–f	Collapse, split by control parameter	The cores fall on the parameter-free unit Gaussian in all 12 conditions and the tails survive. Swelling changes nothing (e), while precursor mass (d) and temperature (f) change the tail. Showing one panel per control parameter answers “which parameter matters?” at a glance.
g	Universal onset	With the Gaussian divided out, all 12 curves sit at 1 and take off at the same Δ^* . This is the most direct visual evidence for a fixed crossover energy.
h	Quantitative summary	What is invariant ($w \approx 1$, Δ^*) and what is not (λ , $E_{3\sigma}$), with error bars and the replicate spread visible.
i	Why pooling is necessary	With 60–180 frames, a single trajectory cannot show the tail reliably even if every junction has it. This answers the referee who asks why individual junctions “look Gaussian”.

Included, moved and dropped

Item	Decision	Reason
Excess ratio $\hat{P}/G$ (g), $E_{3\sigma}$ metric, energy axis on f, null-model panel (i), split-half test (c)	Added	Each answers a specific referee question with one plot. $E_{3\sigma}$ is more robust than λ , because it does not depend on a fit window.
Two-state fits as the lines in d–f	Main	They pass smoothly through the crossover and yield ϵ and s^2 , which predict Δ^* (Section 5 draft, paragraph 6). The piecewise exponential lines from Origin are replaced; λ is still reported in h.
4-arm PEG-SH control	Main (d, h)	A network of different topology is the closest thing to a mechanism test in the present data. Confirm its chemistry and strand mass.
Force vs δx in fN (Origin force plots)	SI	It is a derivative of the same histograms and adds no new data. The pN numbers go in the text; the concept is the inset in a.
Individual Gaussian and non-Gaussian junction histograms	SI	13-bin, two-decade histograms chosen by hand look cherry-picked. Use a random gallery in the SI instead, and panel i in the main figure.
$\pm$ branches and x/y overlay; lag-time invariance of $P(\Delta r)$; MSD plateau	SI or §3	These are prerequisites (symmetry, stationarity) rather than the result.
Raw trajectories (jet colour map)	SI, re-coloured	Nature rules: use viridis for time.
Glass or jamming comparison panel	Dropped	This belongs in the discussion. The temperature trend runs opposite to activated cage escape, so a panel would invite the wrong analogy.

Data-consistency flags (read before finalising)

- σ_i distribution vs pooled width.** For PEGDA 5 kDa (2 h), the σ_i plot gives a mean of 12.5 nm (s.d. 5.6 nm, range 1.5–25 nm), but the Gaussian fit to the pooled $P(\delta x)$ of the same sample gives 9.4 nm. A pooled variance cannot be smaller than $\langle \sigma_i^2 \rangle = (13.7 \text{ nm})^2$. A mixture of Gaussians with the plotted σ_i would also have *heavier* pooled tails than measured. Possible causes: σ_i is a 2D or

radial quantity, it includes localisation error, or the two plots come from different gels or pipelines. $\hat{P}$ depends on σ_i , so resolve this first.

2. **Same sample, two answers.** PEGDA 20 kDa (2 h) appears in the mass series ($\lambda = 1.24$, $E_{3\sigma} = 10.3$) and in the swelling series ($\lambda = 0.77$, $E_{3\sigma} = 5.9$). The spread between gels is therefore about $\times 1.7$, and the " λ rises with mass" claim rests on the 5 kDa vs 10–20 kDa difference. Add replicate gels ($n \geq 3$) before stating a mass trend beyond 10 kDa.
3. **Numbers from the previous draft are not reproduced.** $\lambda = 0.65 \rightarrow 1.10$ and $F = 0.50 / 0.23$ pN do not match any plot in the deck. The Origin exponential lines give $\lambda = 0.60 / 0.69 / 0.90$ (mass) and $0.80 \rightarrow 0.55$ ($5 \rightarrow 45$ °C). Log-space, count-weighted fits over $|\Delta| \geq 2.5$ give $0.63 / 0.80 / 1.24$ and $0.84 / 0.79 / 0.69 / 0.85 / 0.54$. The difference comes from the fit window and weighting; a linear-scale least-squares fit overweights the crossover region. Use maximum likelihood on the raw $|\Delta| > \Delta^*$ values, and quote the window. Your own force plots give plateaus of 0.75, 0.54 and 0.46 pN (5, 10, 20 kDa). These agree with $F \approx 1.5 k_B T / \sigma$ using the pooled σ . Use 0.5–0.75 pN.
4. **Δ^* is not strictly independent of temperature.** It rises from 2.52 to 2.85 ($5 \rightarrow 45$ °C) and U^* from 3.2 to 4.1 $k_B T$. Each point is within about 1.5 s.e., but the trend is monotonic. Tail weight falls with temperature ($E_{3\sigma}$ from 8.7 to 3.3). Write "nearly constant" and report the trend. The statement " Δ^* is unchanged by temperature, so the cost is entropic" needs revising (Section 5 draft, paragraph 7).
5. The temperature series was swollen for 1 h and the others for 2 h. Say so in the legend (done).
6. **Placeholders:** the split-half values in c are simulated from the measured σ_i ; replace them with the real split-half σ_i . The "this study" band in i is an estimate from the trajectory plots (about 180 frames) and the MSD (0.06–10 s); replace it with the actual distribution of N . Then run the same Shapiro–Wilk test on the real trajectories and plot the measured fraction on the panel. The simulated curves use the real fitted $\hat{P}$.
7. All numbers were digitised from the Origin EMF files in the deck (data/), so they carry about 1% digitisation error. Re-run `make_fig5.py` on the raw histograms before submission.

Format

183 mm $\times$ 162 mm (double column), Arial 5–7 pt, panel letters 8 pt bold, 0.5 pt axes with inward ticks on a closed box, Okabe–Ito colours with a cold-to-warm order for temperature. Vector PDF and SVG, with fonts kept as text. In this build Liberation Sans, a metric-identical clone of Arial, stands in; on a machine with Arial, re-running the script embeds Arial. Keep the colours for 5/10/20 kDa and 4-arm fixed across all figures.

5. Cooperative excursions beyond the harmonic cage

Draft. [] marks numbers or statements to check against the raw data.

Section 3 showed that each junction fluctuates about a fixed mean position in a Gaussian cage whose width σ_j differs from junction to junction. Pooling the displacements of many junctions therefore mixes cages of different widths, and the pooled distribution $P(\delta x)$ has tails that no single Gaussian describes (Fig. 5b). To separate these tails from the static heterogeneity, we rescale each displacement by the width of its own junction, $\Delta = \delta x / \sigma_j$, and pool about 10^6 rescaled displacements per condition. The rescaling is justified because σ_j is a property of the junction and not of the sampling. Widths measured from the first and second halves of the same trajectory agree ($r = [0.97]$), while widths of different junctions vary by a factor of about three (Fig. 5c).

After rescaling, the core of the distribution $\hat{P}(\Delta)$ falls on the unit Gaussian, with no adjustable parameter, in all twelve conditions we measured: three PEGDA precursor masses and a four-arm control, three swelling times and five temperatures (Fig. 5d–f). A Gaussian fit to $|\Delta| < 2$ gives a width of 1.03 ± 0.02 (Fig. 5h). Up to about two cage widths, every junction is described by the phantom harmonic cage of Section 3 alone.

Beyond the core, $\hat{P}$ decays far more slowly than the Gaussian. Displacements larger than $3\sigma_j$ are 3–10 times more frequent than a harmonic cage allows ($E_{3\sigma}$ in Fig. 5h). Over the range we resolve, the tail is close to exponential, $\hat{P} \propto \exp(-|\Delta|/\lambda)$, with $\lambda = 0.5$ – 1.2 . The departure begins at the same point in every condition. The ratio of $\hat{P}$ to the unit Gaussian G stays at 1 up to $|\Delta| \approx 2$ and reaches 2 at $\Delta^* = 2.63 \pm 0.11$ (mean $\pm$ s.d. over 12 conditions; Fig. 5g). In energy units, $U = -k_B T \ln \hat{P}$, the cage is harmonic up to $U^* = \frac{1}{2} \Delta^{*2} \approx 3.5 k_B T$ and then grows roughly linearly (Fig. 5a). Equivalently, the restoring force rises linearly inside the cage and saturates beyond Δ^* at about $1.5 k_B T / \sigma_j$. For the pooled cage widths this is 0.75, 0.54 and 0.46 pN for the 5, 10 and 20 kDa networks (Fig. 5a, inset; Supplementary Fig. [X]).

The three control parameters act differently on the tail. Swelling for 0.5, 2 or 24 h leaves the whole distribution unchanged, both the onset and the tail (Fig. 5e). The core behaves the same way: its width σ does not depend on swelling (Section 3), as expected for phantom junctions, whose fluctuations do not depend on macroscopic deformation [James & Guth]. Precursor mass and temperature change the weight of the tail but barely move its onset. $E_{3\sigma}$ increases from 3.2 for 5 kDa to 7.6 and 5.9–10.3 for 10 and 20 kDa. It decreases on heating from 8.7 at 5 °C to 3.3 at 45 °C, while Δ^* shifts only from 2.52 to 2.85 (Fig. 5f,h). The four-arm PEG-SH network has a lighter tail ($E_{3\sigma} = 4.2$) than PEGDA of comparable strand length [10 kDa between junctions, assuming 5 kDa arms]. The same nominal 20 kDa sample measured on two independent gels gives $E_{3\sigma} = 10.3$ and 5.9. Differences smaller than about a factor of 1.7 between gels are therefore not yet significant [replicates].

No single-strand effect produces this shape. For Gaussian strands the phantom cage is harmonic at every displacement. Finite extensibility and good-solvent statistics make strands stiffer at large extension, which has the wrong sign, and excluded volume softens them uniformly, which has the wrong shape. The softening must come from what the strands are attached to. We propose that large excursions are cooperative. Most of the time the junction sits in its phantom cage (the closed state), and moving it means stretching its own f strands against fixed neighbours. Occasionally the neighbourhood rearranges as a whole: neighbouring junctions move in the same direction, and slack stored in nearby loops or dangling ends is released (Fig. 5a). While this lasts the cage is open. The strands between junctions that move together barely stretch, so the junction pays much less per unit displacement. The simplest form of this picture is a two-state cage,

$$\hat{P}(\Delta) = (1 - p) G(\Delta; 1) + p G(\Delta; s), \quad (1)$$

where $G(\Delta; s)$ is a normalised Gaussian of width s , p is the fraction of time (or of junctions) in the open state and s is the width of the open cage in units of σ_j . A distribution of open-cage widths would turn the Gaussian tail of the second term into an exponential one, which is the usual route to Laplace tails in heterogeneous diffusion [Chubynsky & Slater 2014; Chechkin et al. 2017]. Within our dynamic range a single width suffices.

Equation (1) describes all twelve conditions (lines in Fig. 5d–f), with $p = 0.05$ – 0.14 and $s = 1.7$ – 2.7 for the PEGDA networks. The two parameters are correlated, but two combinations are well determined and have a direct meaning. First, the open cage is $s^2 \approx 3$ – 7 times softer than the closed one. Written as springs in series, $1/k_{\text{open}} = 1/k_{\text{strands}} + 1/k_{\text{env}}$, this means the rearranging neighbourhood is 2–7 times (typically 3–4 times) more compliant than the junction's own strands. Second, the cost of opening the cage is $\epsilon = \ln s + \ln[(1 - p)/p] = 3.0 \pm 0.4 k_B T$ across the PEGDA conditions. The model predicts the crossover from these two quantities, $\Delta_c^2 = 2\epsilon s^2 / (s^2 - 1)$, giving $\Delta_c = 2.6$ – 3.0 against the measured $\Delta^* = 2.5$ – 2.85 . The crossover is thus the displacement at which stretching the junction's own strands further costs more than rearranging its neighbourhood. It is set by an energy of a few $k_B T$, which is why Δ^* is nearly constant in units of σ_j .

Each invariance constrains what rearranges. Insensitivity to swelling, which dilutes the network [about threefold], rules out contacts or entanglements between interpenetrating strands. It points instead to connectivity: loops, dangling ends and neighbouring junctions. This is the phantom limit of the constrained-junction picture [Flory 1977; Flory & Erman 1982]. Longer precursors give heavier tails, consistent with more stored slack per junction. The four-arm network's lighter tail is consistent with its lower density of loops and dangling ends [Sakai et al. 2008; Zhong et al. 2016] [confirm chemistry]. Temperature is the most informative parameter. If large excursions were thermally activated escapes from the cage, as in supercooled liquids, they would become more frequent on heating. We observe the opposite. In the two-state fit ϵ shows no systematic trend with temperature (2.6–3.6 $k_B T$), and the open cage is narrowest at 45 °C ($s = 1.7$, against 2.1 at 5

°C), although s is not monotonic in between. We tentatively attribute this to the decrease in the solvent quality of water for PEG on heating, which leaves less slack available for rearrangement [test: $\sigma(T)$ and $\ell = \lambda\sigma(T)$].

Most individual trajectories nonetheless look Gaussian, and this is expected. With [60–180] frames per junction, a trajectory samples only a few displacements beyond Δ^* . Trajectories of this length drawn from the pooled $\hat{P}$ are flagged as non-Gaussian by a Shapiro–Wilk test in only a quarter to a half of cases (median 25–50%; 10–85% across conditions), even though every one of them has the pooled tail. For Gaussian junctions the figure is 5% (Fig. 5i). A single short trajectory without a visible tail therefore does not identify a Gaussian subpopulation. Applied to our trajectories, the same test flags [x]% of junctions, [consistent with / below] the homogeneous prediction. Two further tests separate junctions that switch between states from a subpopulation with permanently open cages. One is the clustering in time of large excursions within a trajectory; the other is whether the tail persists in the longest trajectories (Supplementary Fig. [X]).

A Gaussian core with exponential tails is the signature of cage breaking near the glass and jamming transitions [Chaudhuri, Berthier & Kob 2007; Weeks et al. 2000] and of "Brownian yet non-Gaussian" diffusion [Wang et al. 2012]. In those systems it is usually traced to a distribution of local mobilities. The analogy has limits here. Our displacements are bounded and stationary, and the tail weakens rather than grows with temperature. What this system adds is a harmonic baseline known with no free parameter, so the cooperative part can be isolated as an excess over the phantom cage and measured directly. In mechanical terms the open cage is a soft local mode of a disordered, nearly isostatic network [Wyart 2005; Zaccone & Scossa-Romano 2011]. The series-compliance form above connects the two descriptions (Supplementary Note [X]).

Editorial answers to questions 2–8

Q2. Should we show the restoring force (spring-like, then flat)?

Not as its own main panel. $F = -k_B T \ln \hat{P}/d\delta x$ contains exactly the information in the semilog $\hat{P}$ and is noisier, being the derivative of a histogram. A referee would count it as the same data twice. It does add two things worth keeping. One is a picture: the cage yields, and beyond Δ^* the force stops rising. The other is a number in physical units: the force saturates at about $1.5 k_B T/\sigma_j$, which is 0.75 / 0.54 / 0.46 pN for 5 / 10 / 20 kDa, taken from your own force plots. So: a small inset in panel a (done), the number in the text, and the fN/pN plots in the SI. Replace both the 10–100 fN of V2 and the 0.50 / 0.23 pN of the previous draft.

Q3. Fit core and tail together? Does a good fit convince? What do the parameters mean?

Fit, but do not use the quality of the fit as the argument. Over three decades, a two-Gaussian mixture, a Gaussian plus exponential, and a superstatistical model all fit, and referees know this. The fit earns its place in three ways:

- **It gives interpretable numbers.** $p = 0.05$ – 0.14 is the open-state fraction. $s = 1.7$ – 2.7 means the open cage is $s^2 \approx 3$ – 7 times softer. In series-spring form, the neighbourhood is 2–7 times (typically 3–4 times) more compliant than the junction's own strands during a rearrangement.
- **It makes a prediction you can check.** The opening cost $\varepsilon = \ln s + \ln[(1-p)/p]$ is $3.0 \pm 0.4 k_B T$ for all PEGDA conditions. The crossover implied by (p, s) , $\Delta_c = [2\varepsilon s^2/(s^2 - 1)]^{1/2} = 2.6$ – 3.0 , matches the model-free $\Delta^* = 2.5$ – 2.85 . This consistency between two independent readings of the same curve is more persuasive than any χ^2 .
- **Global fits test the invariance.** Fit all conditions with a shared Δ^* (or ε) and compare against free parameters by AIC. This turns "looks the same" into a number.

Caveat: p and s are anticorrelated. In the main text quote ε , s^2 and the model-free $E_{3\sigma}$, and put the (p, s) table with its covariance in the SI.

Q4. Harmonic width depends on mass and temperature but not swelling, and so does the rescaled tail. What does this mean, and do we show it?

Yes, show it. It is the strongest mechanistic evidence you have, which is why panels d–f are split by control parameter.

- **σ independent of swelling** is the phantom-network prediction: junction fluctuations do not depend on macroscopic strain (James–Guth). A constrained-junction network (Flory–Erman) would instead show σ growing on swelling as the constraints from neighbours relax. This result belongs to §3.
- **Tail independent of swelling** means the cooperative mode is set by connectivity, not by density or contacts, which swelling dilutes. It is topological.
- **Tail depends on mass:** longer strands carry more stored slack per junction. The 4-arm control supports a topological origin; confirm its strand mass. Caveat: the 20 kDa replicate spread is about $\times 1.7$.
- **Tail depends on temperature, in the "wrong" direction for activated hopping.** Tail weight *falls* on heating, which argues against glass-like thermally activated escape. Your earlier line " Δ^* unchanged by T, so the cost is entropic" should become " $\varepsilon \approx 3 k_B T$ with no systematic trend in temperature, while the tail weight falls". The fits hint that the open cage narrows on heating, but s is not monotonic. Measure $\sigma(T)$ and check whether $\ell = \lambda\sigma$ is independent of temperature; if so, the tail has a fixed physical length (slack) in nm.

Q5. Show both branches or only the positive half?

Both, once, where it matters: the unscaled pooled plot (b) shows $\pm\delta x$. The symmetric "tent" against the Gaussian parabola is instantly readable and proves symmetry. For the rescaled analysis fold to $|\Delta|$ (d–g). This doubles the statistics, halves the clutter and is standard practice, provided you say so. In the SI, overlay $+/-$ and x/y with an asymmetry measure, for example $|P(\Delta) - P(-\Delta)|/d\Delta$.

Q6. About 10% of single junctions are non-Gaussian: is "too few frames" a legitimate explanation? Show single junctions?

The claim is legitimate, but it must be *demonstrated*, not asserted, and this is what panel i does. With 60–180 frames, even if every junction had exactly the pooled $\hat{P}$, a Shapiro–Wilk test would flag only 25–50% of junctions (median; 10–85% across conditions), against 5% for truly Gaussian junctions. Apply the same test to your trajectories and plot the measured fraction on the panel:

- On the homogeneous curve: the tail belongs to every junction (switching).
- Well below it, but above 5%: a subpopulation with a tail.
- Also check that the flagged fraction rises with N when you split junctions by trajectory length.

If your "10%" came from inspecting per-junction histograms by eye, that is a weaker test than Shapiro–Wilk, and the two numbers are not directly comparable. Do not put a hand-picked Gaussian junction next to a hand-picked non-Gaussian one in the main figure; it reads as cherry-picking. Use a random gallery with the pooled $\hat{P}$ overlaid, in the SI.

Q7. A schematic to lead the figure?

Yes: one schematic, in panel a, and make it do work. Pair the cartoon of the two cages with the measured landscape, so the reader sees the hypothesis and the evidence together. Keep the cartoon minimal (junction, four strands, co-moving neighbours). Detailed loop and dangling-end drawings belong to the mechanism section, or to the SI if §6 is folded into §5.

Q8. Justify $\Delta = \delta x / \sigma_i$ up front: how?

Three checks, which the figure now covers:

1. σ_i is persistent: split-half reproducibility (c).
2. σ_i is heterogeneous: its distribution (c, inset).
3. Rescaling collapses every core onto the unit Gaussian with no free parameter (d–f; $w = 1.03 \pm 0.02$ in h).

Two methodological points a sharp referee will raise:

- Estimate σ_i robustly (a core fit or $1.4826 \times \text{MAD}$), not as the full s.d. A full s.d. absorbs the tail into σ_i and shrinks the very excess you report.
- Cross-validate by rescaling each half-trajectory with σ_i from the other half. This removes any circularity; if $\hat{P}$ is unchanged, the rescaling is not manufacturing the tail.

Finally, resolve flag 1 (σ_i vs pooled σ) before anything else.